

during the first nationwide Indian lockdown, as economic headwinds led to SIM consolidation and shock to the smartphone supply chain impacted 4G adoption. Since then, broadband connections have continued their relentless march according to TRAI, increasing month by month to hit 792.8 million at the end of June, with broadband teledensity increasing from 85.9% in June 2020, to 88.1% in June 2021.

Overall teledensity levels don't tell the full story though. A recent Deloitte report highlighted that the digital divide between urban and rural India remains stark, while fixed broadband penetration remains very low in the market. TRAI recently submitted a wide range of recommendations to the DoT, as part of what it calls a 'Roadmap to Promote Broadband Connectivity and Enhanced Broadband Speed.' These include a change to the definition of broadband, increasing the minimum speed from 512 Kbps to 2 Mbps, and introducing definitions for speed categories of basic, fast and super-fast for speeds beyond that. In addition to a slew of other measures designed to help improve the nation's broadband infrastructure, TRAI has also recommended adjustments to the way AGR fees are calculated. This is to encourage broadband provision by the nation's numerous cable operators, as well as a pilot scheme to provide subsidies to encourage broadband adoption in rural areas.

Q Was there a difference in internet speeds during the COVID-19 pandemic between urban and rural internet users in India?

Suttles: The pandemic has really underscored how critical access to the internet is in enabling economies to continue functioning while locked down. Nevertheless, there's now a strong focus on the fact that the digital divide is still present in many markets, and India is no different in that respect. In the largely rural state of Meghalaya, median mobile download speeds in Q2 2021 were 5.47 Mbps, but for the 10% of users who typically experienced the slowest speeds, that figure was a mere 0.72 Mbps. However, compare that to Delhi, where median mobile down-

load speeds were 12.43 Mbps, with the lower 10% standing at 2.35 Mbps. It's clear that internet speeds for many users in rural India fall short of TRAI's new 2 Mbps threshold. So it will be interesting to track the impact of its proposed measures on performance going forward.

Q Do you have data on what Indians were using the internet for during the pandemic?

Suttles: While this isn't something we track, it's apparent that stay-at-home orders pushed all of our lives online, from our work to our social relationships and entertainment. I've had to manage Ookla remotely from home since the pandemic started. Like many businesses, we've had to adapt to not being in an office together, and all the challenges that it brings. It's clear that lockdowns have driven an explosion in the use of video calling as people work from home and stay in touch with family and friends. It's also driven a massive increase in video streaming and online gaming. I know I've personally been watching more movies over the weekends. I suspect many people in India had the same experience. Additionally, these are all activities that demand a lot of bandwidth, so it puts the spotlight on operator networks and their ability to manage network congestion. To that end, we recently announced video testing from Speedtest (available on both our iOS and Android apps), allowing users to measure the quality of their video streaming experience.

Q Does Ookla expect this change to remain post the pandemic, or will things go back to pre-pandemic numbers once things return to a semblance of normality?

Suttles: We've already seen internet speeds recover in India following their decline during the first lockdown. I think this points to two things – one, that operators have increased capacity in their networks to handle the increased demand for data, and two, that we've seen demand equalize as the economy opened back up. The longer-term reality will likely be somewhere in the middle. The pandemic has lasted

long enough to ingrain many of these new behaviours, while businesses have invested heavily in order to cater to this changing demand. Think of the slew of streaming services that are now available in the market as an example. On the other hand, as things return to some semblance of normality and travel rebounds, we'll see the impact of the pent-up demand for in-person meetings and social gatherings.

Q Streaming services reduced the bitrate of their stream during the pandemic. How much do you think this helped people with limited monthly bandwidth?

Suttles: Streaming services are constantly looking to improve the bitrate efficiency of their video content, as it directly impacts the quality of experience. Late last year, Netflix announced that it was re-encoding its library to adjust the bitrate based on the type of content being shown, which it said would reduce the average bitrate by a staggering 50%. These kinds of advances, alongside forced reductions in streaming bitrates during the pandemic, have helped to relieve network congestion. At the same time, this also means that anyone on a metered internet service will likely see their monthly allowance go further than before.

Q We saw a lot of service providers offer extra data during the pandemic? Does providing extra data hurt the service provider?

Suttles: We saw operators around the world implement a range of measures to help people stay connected during the pandemic. These measures included providing extra data and upgrading them to faster speeds, migrating to online recharging for pre-paid customers, and even distributing devices to those most impacted by the economic fallout. Providing extra data to customers helped Indian operators cater to the increased level of demand. As a result, it will have helped reduce the potential for churn. It's no secret that data tariffs in India are among the lowest in the world, so any initiatives like this need to be accompanied by increases in tariffs. This will ensure that the sector is sustainable. **Q**